

JORDAN NEWMAN

Distributed Systems Engineer | Compute Infrastructure, Runtime, Systems Software

Marlboro, NJ | 347-739-4731 | phoem@mac.com | github.com/phoem

Selected public repositories are available. The most significant engineering repositories are private; access is available upon request.

SUMMARY

Distributed systems and systems software engineer with more than 20 years of software development and production infrastructure experience. Architected, built, debugged, and operated compute and networking platforms spanning approximately 3,000-4,000 servers, 10 locations, and more than 65 Gbps of peak traffic. Built production runtimes, kernel modules, performance tools, DDoS defenses, telemetry, deployment automation, and Kubernetes-based delivery workflows. Hands-on lab experience includes kubelet, containerd, runc, and NRI plugin architecture. Deeply familiar with Linux kernel architecture and systems internals, backed by production FreeBSD kernel-module development.

TECHNICAL SKILLS

- **Languages:** C, C++, Go, Python, C#, Objective-C, PHP, Perl, JavaScript, Node.js, Shell, Visual Basic
- **Systems and Performance:** Linux kernel architecture, FreeBSD kernel modules, system calls, processes, memory, networking, filesystems, scheduling, kqueue, sendfile(), GDB, performance debugging, troubleshooting, non-blocking I/O
- **Networking:** TCP/IP, IPv4, sockets, HTTP, DNS, CDN architecture, packet capture, libpcap, IPFW, host networking, authentication, telemetry
- **Cloud and Orchestration:** Kubernetes, Docker, Terraform, Azure DevOps, CI/CD, security scanning, deployment automation; experimental kubelet, containerd, and runc; NRI plugin fundamentals
- **Databases/Web:** MySQL, MultiValue/Universe, OAuth/OAuth2, SMTP OAuth, React, jQuery, PHP, web services

SELECTED TECHNICAL HIGHLIGHTS

- Designed and deployed PrimeHTTPD, a custom event-driven compute and HTTP runtime across approximately 200 production FreeBSD servers.
- Supported more than 150,000 concurrent connections with kqueue, persistent connections, non-blocking sockets, zero-copy sendfile(), and TCP optimizations.
- Removed runtime bottlenecks by isolating blocking disk work in dedicated I/O workers and using SF_NODISKIO-aware handling.
- Architected and operated distributed compute, hosting, and CDN infrastructure across thousands of servers and multiple regions.
- Built observability, traffic analysis, DNS monitoring, authentication, deployment, security, and DDoS mitigation systems for 24/7 operations.
- Implemented Kubernetes, Docker, Terraform, Azure DevOps, CI/CD, and security-scanning workflows across engineering teams.
- Used kubelet, containerd, and runc in hands-on experiments and studied NRI plugin development fundamentals; this was exploratory, not production runtime development.
- Applied strong Linux kernel knowledge alongside production FreeBSD kernel development, without claiming Linux kernel-module authorship.

SELECTED SYSTEMS PROJECTS

- **PrimeHTTPD:** Built a non-blocking, kqueue-based HTTP/CDN runtime in C for FreeBSD. Used sendfile(), SF_NODISKIO, TCP_NODELAY, TCP_NOPUSH, O_NONBLOCK, accept filters, persistent connections, and dedicated I/O workers.
- **VirtualDir:** Developed a production FreeBSD kernel module that intercepted filesystem-related syscalls and remapped paths from configuration; included the vdcli runtime management tool.
- **PrimeDump:** Built a libpcap-based network troubleshooting tool that decoded Ethernet, IPv4/IPv6, TCP, UDP, and ICMPv6 traffic with ncurses views and IPFW integration.
- **PrimeDNSTop:** Built a DNS traffic monitor that parsed UDP/53 and RFC 1035 queries to detect recursion attacks and rank source and domain activity.
- **TAFOS:** Developed an educational x86 operating-system kernel in C and assembly with a custom MBR bootloader, protected mode, IDT, heap allocator, port I/O, VGA output, and GDB support.

PROFESSIONAL EXPERIENCE

Advantive - Senior Development Specialist

Remote / Tampa Bay, FL | July 2022 - Present

- Implemented Kubernetes, Docker, Terraform, Azure DevOps, CI/CD, and security-scanning workflows across engineering teams.
- Collaborated on the Architecture Team to design solutions, define standards, guide platform direction, and make decisions amid evolving requirements and competing priorities.
- Developed shared libraries, technical documentation, and AI-assisted engineering workflows with architects and engineering teams.

DDI Systems - Senior Development Specialist

Manalapan, NJ | November 2021 - July 2022

- Modernized legacy VB.NET components in C# and integrated Microsoft Office 365 SMTP OAuth for secure customer-facing workflows.

ISPrime LLC - CEO

Weehawken, NJ | April 2018 - January 2020

- Led datacenter modernization, operational restructuring, and improvements to internally developed monitoring, security, and infrastructure platforms.

DDI Systems - Computer Programmer

Manalapan, NJ | April 2017 - April 2018

- Built ERP integrations and automated order, catalog, image, SDS, SFTP, SellerCloud, Essendant, and MultiValue/Universe data workflows.

Too Much Media LLC - Computer Programmer

Morganville, NJ | January 2016 - April 2017

- Developed real-time browser chat integrated with ticketing systems and implemented OAuth 2.0 authorization for protected sites.

MFCXY, Inc. - Computer Programmer

Chicago, IL | December 2014 - September 2015

- Developed backend and Windows client features and middleware for cross-platform database migration.

AJPM, LLC - Programmer / Owner

New Jersey / Remote | November 2012 - December 2013

- Built automated purchasing, inventory, repair, resale, and financial workflow software.

ISPrime Inc. - CIO / Partner

Weehawken, NJ | March 2001 - December 2014

Infrastructure Leadership

- Architected and operated compute, hosting, and CDN infrastructure spanning approximately 3,000-4,000 servers, 10 locations, multiple datacenters, and more than 65 Gbps of peak traffic.

Systems Software Engineering

- Designed and built PrimeHTTPD, a non-blocking HTTP/CDN runtime in C for FreeBSD; deployed it across approximately 200 servers and supported more than 150,000 concurrent connections.
- Debugged performance and operational issues across the runtime and operating-system boundary using sendfile(), non-blocking I/O, accept filters, persistent connections, and dedicated I/O workers.
- Developed production FreeBSD kernel modules including VirtualDir, KeepClean, and StatCache.

Security and Networking

- Built DDoS mitigation, packet and DNS analysis, monitoring, telemetry, authentication, deployment, and security software for production infrastructure.

FatWallet.com - Systems Administrator

Wisconsin / Remote | December 2000 - June 2001

- Rebuilt production infrastructure to resolve performance and stability issues without customer-facing downtime and created automated availability and health alerting.

EDUCATION

Brookdale Community College - Computer Science coursework; left during first semester to pursue ISPrime full-time during rapid company growth.